

# DS Unit 2 Question Bank — Stacks & Queues

Data Structures (DS) · Unit 2 · Stacks & Queues · 2025

---

Important and frequently asked questions, grouped by marks.

## Part A — Short answer (2 marks)

1. What is meant by Stack ADT? Discuss its advantages and limitations.
2. Illustrate Infix to Postfix Conversion with a worked example.
3. What is meant by Recursion and the Call Stack? Discuss its advantages and limitations.
4. What is meant by Queue ADT? Discuss its advantages and limitations.
5. Compare Deque and Priority Queue with Infix to Postfix Conversion. When would you prefer one over the other?
6. Write short notes on Applications of Stacks and Queues.

## Part B — Long answer (10 marks)

1. Compare Stack ADT with Deque and Priority Queue. When would you prefer one over the other?
2. Define Infix to Postfix Conversion. Derive or justify the key properties involved.
3. Discuss the role of Recursion and the Call Stack in Stacks & Queues.
4. Illustrate Queue ADT with a worked example.
5. Write short notes on Deque and Priority Queue.
6. Write short notes on Applications of Stacks and Queues.
7. Compare Stack ADT with Infix to Postfix Conversion. When would you prefer one over the other?
8. Discuss the role of Infix to Postfix Conversion in Stacks & Queues.
9. Illustrate Recursion and the Call Stack with a worked example.
10. Illustrate Queue ADT with a worked example.
11. What is meant by Deque and Priority Queue? Discuss its advantages and limitations.
12. Discuss the role of Applications of Stacks and Queues in Stacks & Queues.